Trader Performance vs Market Sentiment Analysis

Methodology

The analysis combines Bitcoin market sentiment data with historical trading data to understand how trader behavior and performance vary across different market conditions. Both datasets were cleaned, timestamps were standardized, and merged on a daily basis.

Key metrics were engineered, including win rate, trade size (USD), trade frequency, daily PnL per trader, and long/short ratio. Since leverage data was unavailable, trade size was used as a proxy for risk exposure. A drawdown proxy was also created using average losses.

Key Findings

1. **Performance varies significantly with sentiment**
   Traders achieve the highest profitability and win rates during Extreme Greed, while performance drops during Fear and Extreme Fear phases. However, win rates remain below 50% across all conditions, indicating that profitability is driven by larger winning trades rather than accuracy.

2. **Aggressive behavior does not improve outcomes**
   During Fear phases, traders increase both trade frequency and position sizes, yet performance does not improve. In contrast, during Extreme Greed, traders reduce position sizes and activity while achieving better results. This suggests that aggressive behavior in uncertain markets leads to inefficiency.

3. **Overtrading reduces profitability**
   Frequent traders consistently underperform compared to infrequent traders, especially during Greed phases. Higher activity does not translate into better outcomes, indicating that selective participation is more effective than continuous trading.

4. **Position sizing is a key driver of profitability**
   High trade size traders generate significantly higher PnL, but their win rates are not proportionally higher. This shows that profitability is more dependent on trade size and payoff distribution than on prediction accuracy.

Strategy Recommendations

1. **Reduce risk exposure during Fear phases**
   Traders should lower position sizes and avoid aggressive trading during Fear and Extreme Fear conditions, as these periods show higher downside risk and lower profitability.

2. **Prioritize selective trading over high frequency**
   Traders should limit trade frequency and focus on high-quality opportunities. Infrequent traders consistently outperform frequent traders without relying on higher win rates.

Conclusion

The analysis shows that market sentiment strongly influences trader behavior, but disciplined execution is the key determinant of performance. Traders who control risk and avoid overtrading achieve better outcomes than those who react aggressively to market conditions.